

2012-2013 – Master 2 – Macro I

Lecture notes #10 : The Hosios efficiency conditions

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Version 1.1
13/11/2012

Changes from version 1.0 are in red

Disclaimer

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1. Can unemployment be efficient ?

Two inefficiencies

- ▶ In matching models, unemployment is a productive input into the creation of new jobs. Thus there is no presumption that the optimal unemployment level is zero.
- ▶ To have a low unemployment rate, vacancies must be high. But vacancies consume resources, so that potentially we can have too many vacancies.
- ▶ On the other hand, the unemployed are also costly because they do not produce. It is also possible to have too much unemployment.
- ▶ There are two central market failures in the matching model : congestion externalities and appropriability problems.

1. Can unemployment be efficient ?

Congestion externalities

- ▶ The congestion externalities are as follows. An increase in search exerts positive externalities on the other side of the market. Unemployment reduces the length of vacancies ; conversely vacancies reduce the length of unemployment
- ▶ Because $m()$ is concave, search activity exerts a negative externality on the same side of the market : More unemployment reduces the job finding rate.

1. Can unemployment be efficient ?

Appropriability

- ▶ A surplus arises from turnover costs, and part of it surplus is appropriated by the worker in the bargaining process.
- ▶ The firm only appropriates a fraction of the surplus created by a vacancy, but pays the entire vacancy cost.
- ▶ Hence the incentives to post vacancies are too low.
- ▶ This is the hold-up problem discussed by Grout (Econometrica, 1984).
- ▶ A symmetrical issue arises on the worker's side when evaluating the value of being unemployed :
- ▶ only a fraction of the value of future jobs will be appropriated, hence the value of looking for a job (=opportunity cost of working now) is underestimated.

2. Computing the optimum

Market versus Social Planner

- ▶ Market allocations (W, θ) given by :
 - ▶ A job creation condition

$$\frac{c}{q(\theta)} = J = W(1 - \varphi)$$

- ▶ An evolution equation for the joint match surplus :

$$rW = y - \theta q(\theta) \varphi W - sW + \dot{W}.$$

- ▶ The social planner's problem is to maximize the PDV of output net of vacancy costs :

$$\max_{\{v_t\}} \int_0^{+\infty} (y(1 - u_t) - cv_t) e^{-rt} dt,$$

subject to the law of motion of unemployment :

$$\dot{u}_t = -m(u_t, v_t) + s(1 - u_t).$$

2. Computing the optimum

Market versus Social Planner

$$\begin{aligned} \max_{\{v_t\}} \quad & \int_0^{+\infty} (y(1 - u_t) - cv_t)e^{-rt} dt, \\ \text{s.t.} \quad & \dot{u}_t = -m(u_t, v_t) + s(1 - u_t). \end{aligned}$$

- ▶ state variable u , control variable v
- ▶ The Hamiltonian is

$$H_t = (y(1 - u_t) - cv_t)e^{-rt} + \lambda_t e^{-rt}(-m(u_t, v_t) + s(1 - u_t))$$

- ▶ The state variable is unemployment, its marginal value λ is typically negative.

2. Computing the optimum

Market versus Social Planner

$$H_t = (y(1 - u_t) - cv_t)e^{-rt} + \lambda_t e^{-rt}(-m(u_t, v_t) + s(1 - u_t))$$

- ▶ The first order conditions are

$$\frac{\partial H}{\partial v} = 0 \iff c = -\lambda_t \frac{\partial}{\partial v} m(u, v).$$

$$\frac{\partial H}{\partial u} = (-\dot{\lambda} + r\lambda)e^{-rt} \iff -y - \lambda \frac{\partial m}{\partial u} - s\lambda = r\lambda - \dot{\lambda}$$

- ▶ The co-state variable λ is the marginal social value of an extra unemployed worker.
- ▶ Let $\mu = -\lambda$. μ is the marginal social value of one extra job.

2. Computing the optimum

Market versus Social Planner

- FOC rewrite

$$c = \mu \frac{\partial}{\partial v} m(u, v).$$

$$r\mu = y - s\mu - \mu \frac{\partial m}{\partial u} + \dot{\mu}$$

- the second equation is the standard asset valuation equation. The dividend of an additional job is output y minus the value of the jobs that the worker's search activity would have created should he be unemployed instead. This value is $\mu \frac{\partial m}{\partial u}$. The capital gains are $-s\mu + \dot{\mu}$.

2. Computing the optimum

Market versus Social Planner

- ▶ By comparing the preceding formulas between equilibrium and optimum, we understand the market failures.

2. Computing the optimum

The job creation decision

- ▶ The market equilibrium condition is

$$c = q(\theta)W(1 - \varphi).$$

- ▶ That is, vacancy cost $[c]$ = probability of filling a vacancy $\times$ value to the firm of a filled job = probability of filling a vacancy $[q(\theta) = \frac{m(u,v)}{v}] \times$ value to the match of a filled job $[W] \times$ fraction of that value going to the firm $[1 - \varphi]$.
- ▶ The socially optimal job creation condition is

$$c = \mu \frac{\partial}{\partial v} m(u, v). \quad (1)$$

That is, vacancy cost = marginal effect of extra vacancy on jobs created $[\frac{\partial}{\partial v} m(u, v)] \times$ marginal social value of a job $[\mu]$.

2. Computing the optimum

The job creation decision

- ▶ The private firm only cares on the *average* probability of filling a vacancy, $\frac{m(u,v)}{v}$ in making its decision, while the social planner cares about the marginal increment in the number of matches generated by an extra vacancy, $\frac{\partial}{\partial v} m(u, v)$. Because of decreasing marginal returns, we have that $\frac{\partial}{\partial v} m(u, v) < \frac{m(u,v)}{v}$. This is the congestion externality at work and it leads to too many vacancies.
- ▶ The social planner considers the whole social value of a job μ , while the private firm only internalizes a fraction $1 - \varphi$ of the private value of the match created by its vacancy. This is the appropriability problem and it leads to too few vacancies.

2. Computing the optimum

The asset valuation equation

- ▶ The evolution equation for W is

$$rW = y - \theta q(\theta) \varphi W - sW + \dot{W}.$$

- ▶ Private opportunity cost of labor is $\theta q(\theta) \varphi W = \frac{m(u,v)}{u} \varphi W$, while the social opportunity cost is $\mu \frac{\partial m}{\partial u}$.
- ▶ The evolution equation for μ is

$$r\mu = y - \mu \frac{\partial m}{\partial u} - s\mu + \dot{\mu} \tag{2}$$

2. Computing the optimum

The asset valuation equation

- ▶ Private opportunity cost of labor is $\theta q(\theta) \varphi W = \frac{m(u,v)}{u} \varphi W$, while the social opportunity cost is $\mu \frac{\partial m}{\partial u}$.
- ▶ The worker/firm pair considers that if the worker were instead unemployed, he would find a job with a probability equal to the average job finding rate, $\frac{m(u,v)}{u}$, and would appropriate a fraction φ of the private surplus of the job he would find.
- ▶ Social planner considers that if one extra worker were unemployed, he would generate a flow of new jobs equal to the marginal product of labor in the matching function, $\frac{\partial m}{\partial u}$.
- ▶ Furthermore, the full social gain of those extra jobs μ is taken into account by the social planner in computing the social opportunity cost of a job.

2. Computing the optimum

The asset valuation equation

Again we see the two market failures :

- ▶ Workers fail to internalize the fact that should they look for a job, they generate extra jobs at a rate lower than their own job finding probability. This is the congestion externality which would lead to too much worker search, i.e. too much unemployment.
- ▶ Workers only appropriate a fraction φ of the private value of the jobs they find. This is the appropriability problem which leads to too little worker search, i.e. too little unemployment.

3. When is the equilibrium optimal?

Equilibrium

$$c = W(1 - \varphi) \frac{m(u, v)}{v}$$

$$rW = y - \theta q(\theta) \varphi W - sW + \dot{W}$$

Optimum

$$c = \mu \frac{\partial}{\partial v} m(u, v)$$

$$r\mu = y - \mu \frac{\partial m}{\partial u} - s\mu + \dot{\mu}$$

- By looking at these equations, we see that the equilibrium delivers $W = \mu$ and $\theta = \theta^*$ if

$$\frac{\partial m}{\partial u} = \varphi \frac{m(u, v)}{u}$$

and

$$\frac{\partial}{\partial v} m(u, v) = (1 - \varphi) \frac{m(u, v)}{v},$$

- or equivalently

$$\varphi = \frac{u \frac{\partial m}{\partial u}}{m} = \eta_u,$$

$$1 - \varphi = \frac{v \frac{\partial m}{\partial v}}{m} = \eta_v.$$

3. When is the equilibrium optimal ?

- ▶ *The share of workers (firms) in the surplus of a match is equal to the elasticity of the matching function with respect to the corresponding search input.*
- ▶ These conditions are called the **Hosios conditions**.
- ▶ That is, Note that under our assumption of constant returns, one implies the other.
- ▶ The HC mean that the appropriability and congestion problems exactly balance each other.

3. When is the equilibrium optimal ?

- ▶ Conversely, what do we need to have $\theta = \theta^*$?
- ▶ Note that since $m(u, v) = vq(v/u)$, we have $\partial m / \partial u = -\theta^2 q'(\theta)$ and $\partial m / \partial v = q(\theta) + \theta q'(\theta)$.
- ▶ We can eliminate μ between (2) and (1) and note again that the dynamics of θ are unstable.
- ▶ We then conclude that θ^* is constant throughout and satisfies

$$(r + s)c = y(q(\theta^*) + \theta^* q'(\theta^*)) + \theta^{*2} q'(\theta^*)c.$$

- ▶ We already know that θ satisfies

$$(r + s)c = y(1 - \varphi)q(\theta) - \theta q(\theta)\varphi c.$$

- ▶ Using these two formulas, we see that to have $\theta = \theta^*$ we must have

$$y(\varphi q(\theta) + \theta q'(\theta)) = -c\theta(\varphi q(\theta) + \theta q'(\theta)).$$

- ▶ Thus we must have $\varphi = -\frac{\theta q'(\theta)}{q(\theta)} = \eta_u$, which is the Hosios condition.

3. When is the equilibrium optimal ?

- ▶ We note that if $\varphi > \eta_u$, there is too much unemployment :
- ▶ the appropriability problem dominates the congestion externality on the firms' side, and the converse on the workers' side.
- ▶ Both effects push unemployment above the optimum.